MATHEMATICS FOR MACHINE LEARNING

CampusX · Data Science Mentorship Program 2023 · Video 14 of 23

SESSION 14
Conditional Probability & Bayes' Theorem

Joint & Marginal Probability · Conditional Probability · Bayes' Theorem · ML Naive Bayes Foundations

- Joint Probability vs Marginal Probability
- Conditional Probability Formulation
- Deriving Bayes' Theorem
- Prior, Likelihood, and Posterior
- Bayes' Theorem Examples (Medical Testing)
- Naive Bayes Classifier Core Mathematics
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Session Notes — Complete Reference

Playlist: Maths for Machine Learning · Instructor: Nitish Singh (CampusX)

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01 · Joint Probability vs Marginal Probability

Joint Probability: The probability of two events occurring simultaneously: $P(A \cap B)$.

Marginal Probability: The probability of an individual event occurring on its own, regardless of other events: $P(A) = \sum P(A \cap B_j)$.

02 · Conditional Probability Formulation

The probability of event A occurring given that event B has already occurred:

CONDITIONAL PROBABILITY FORMULA

$$P(A \mid B) = P(A \cap B) / P(B) \quad (\text{where } P(B) > 0)$$

03 · Deriving Bayes' Theorem

Since $P(A \cap B) = P(A | B)P(B)$ and $P(A \cap B) = P(B | A)P(A)$, we equate the two to derive Bayes' Theorem:

BAYES' THEOREM FORMULA

$$P(A \mid B) = [\ P(B \mid A) \cdot P(A)\ ] \ / \ P(B)$$

04 · Prior, Likelihood, and Posterior

Bayes' Theorem is interpreted in machine learning as updates of beliefs based on new evidence:

- **$P(A)$ (Prior):** Initial probability of the hypothesis before seeing the evidence.
- **$P(B | A)$ (Likelihood):** Probability of the evidence given the hypothesis.
- **$P(B)$ (Normalizing Constant):** Probability of the evidence.
- **$P(A | B)$ (Posterior):** Updated probability of the hypothesis *after* seeing the evidence.

05 · Bayes' Theorem Examples (Medical Testing)

Suppose a disease affects 1% of the population. A test is 99% accurate (true positive rate) and has a 5% false positive rate.

If a person tests positive, what is the probability they actually have the disease?

Applying Bayes' Theorem yields: $P(\text{Disease} | \text{Positive}) \approx 16.6\%$. Even with a highly accurate test, the low baseline rate (prior) makes a positive result likely to be a false positive.

06 · Naive Bayes Classifier Core Mathematics

Naive Bayes models the probability of class y given features $\mathbf{x} = (x_1, x_2, ..., x_d)$:

NAIVE BAYES FORMULATION

$$P(y \mid x) \propto P(y) \cdot \prod_i P(x_i \mid y)$$

The term "Naive" comes from the assumption that all features are **conditionally independent** given the class label.

07 · Probability Trees & Visual Representation

Probability trees are useful visual tools to decompose conditional probabilities. The branches represent conditional probabilities, and path multiplications yield joint probabilities.

08 · Python Naive Bayes Classification Example

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB

X, y = load_iris(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

clf = GaussianNB()
clf.fit(X_train, y_train)
print(f"Accuracy: {clf.score(X_test, y_test):.4f}")
```

09 · Quick Reference Card

🚀 Bayes' Theorem Cheat Sheet

- **Formula:** $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$.
- **Law of Total Probability:** $P(B) = \sum P(B|A_i)P(A_i)$.
- **Naive Bayes:** Assumes feature independence given the class.